

SHERBROOKE LENNOXVILLE, QC, Canada

WMO: 716110

Lat: 45.3686N

Lon: 71.8233W

Elev: 181

StdP: 99.17

Time Zone: -5.00 (NAE)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.4	-22.3	-29.4	0.3	-24.9	-26.5	0.3	-21.7	10.4	-3.6	9.3	-4.7	1.8	60	0.553

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	29.5	21.5	27.9	20.6	26.6	19.9	23.1	27.3	22.1	26.1	21.1	24.9	4.3	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.7	16.8	25.3	20.7	15.7	24.3	19.8	14.8	23.2	69.3	27.4	65.5	26.2	62.1	24.9	27.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.9	7.8	6.9	-30.7	32.0	3.2	1.0	-33.0	32.7	-34.9	33.3	-36.7	33.8	-39.0	34.6		
			-30.5	24.6	3.6	1.1	-33.1	25.4	-35.1	26.1	-37.1	26.7	-39.7	27.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.3	-9.1	-7.6	-2.4	5.3	12.4	17.2	19.7	18.9	14.9	8.0	1.7	-5.0
	DBStd	11.31	7.28	6.52	6.33	4.72	4.26	3.64	2.90	2.89	4.08	4.60	5.34	6.30
	HDD10.0	2464	592	492	384	155	23	1	0	0	6	96	252	465
	HDD18.3	4564	850	725	642	391	189	63	18	28	117	320	498	723
	CDD10.0	1101	0	0	1	14	98	217	302	277	153	34	4	0
	CDD18.3	158	0	0	0	0	6	29	61	47	14	1	0	0
	CDH23.3	1387	0	0	1	9	89	289	498	363	131	6	0	0
	CDH26.7	284	0	0	0	1	16	68	109	65	26	0	0	0
Wind	WSAvg	2.8	3.1	3.1	3.3	3.4	3.0	2.6	2.4	2.2	2.3	2.9	3.0	3.0
Precipitation	PrecAvg	1095	70	63	71	81	97	101	112	119	89	85	96	95
	PrecMax	1417	194	132	115	183	166	164	195	206	167	202	166	189
	PrecMin	884	18	10	12	36	39	38	53	42	37	23	54	43
	PrecStd	136	36	30	25	33	40	33	43	39	33	38	29	41
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.4	10.1	17.4	24.5	28.8	30.9	31.5	30.5	29.7	24.0	18.9	10.7
		MCWB	8.2	7.0	10.9	15.2	18.7	21.7	23.5	22.5	22.0	17.5	14.8	8.6
	2%	DB	5.8	6.2	11.8	20.5	25.8	28.7	29.4	28.5	26.7	20.7	15.1	7.9
		MCWB	4.3	3.5	7.1	12.5	16.5	20.6	21.8	21.2	20.2	15.4	11.6	6.6
	5%	DB	3.4	4.0	8.7	17.0	23.4	26.7	27.7	26.9	24.5	18.3	12.2	5.2
		MCWB	2.2	1.9	5.0	10.3	15.7	19.5	20.9	20.6	19.0	14.3	9.1	4.1
	10%	DB	1.2	2.1	6.2	13.7	21.0	24.7	26.2	25.4	22.3	15.8	9.5	3.1
		MCWB	0.2	0.6	3.5	8.3	14.5	18.5	20.0	19.7	18.1	12.3	7.4	1.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.2	7.8	13.1	17.1	20.9	23.6	25.0	23.8	22.8	19.0	15.4	9.1
		MCDB	9.0	9.4	16.6	22.7	25.8	28.3	29.9	28.0	27.7	22.6	17.8	10.4
	2%	WB	4.5	4.2	7.8	13.8	18.8	22.1	23.2	22.6	21.4	16.5	12.4	6.7
		MCDB	5.4	5.8	10.4	18.6	23.3	26.7	27.2	26.6	25.1	19.4	14.9	7.7
	5%	WB	2.2	2.1	5.5	11.0	17.2	20.8	22.1	21.5	20.1	14.5	9.7	4.2
		MCDB	3.4	3.7	8.0	14.7	21.4	24.8	26.1	25.2	23.5	17.6	11.6	5.1
	10%	WB	0.2	0.6	3.7	8.9	15.4	19.5	21.1	20.6	18.7	12.8	7.3	2.1
		MCDB	1.2	2.0	6.0	13.2	19.5	23.2	24.9	24.1	21.6	15.5	9.3	2.9
Mean Daily Temperature Range	5% DB	MDBR	10.0	11.0	10.4	10.5	11.7	11.0	10.8	11.1	11.5	9.6	8.1	8.0
		MCDBR	10.5	11.7	13.8	17.0	15.9	14.2	12.6	13.2	14.1	13.8	12.4	8.7
		MCWBR	9.8	9.3	9.3	10.0	8.3	7.2	6.3	7.0	8.0	8.8	9.3	8.2
	5% WB	MCDBR	10.3	10.3	11.8	14.2	12.9	11.8	10.9	11.1	12.2	12.1	11.2	8.4
		MCWBR	9.9	8.7	8.7	9.4	7.7	6.9	6.1	6.4	7.5	8.5	9.2	8.2
Clear-Sky Solar Irradiance	taub	0.265	0.285	0.315	0.350	0.374	0.416	0.422	0.401	0.368	0.329	0.306	0.277	
	taud	2.372	2.282	2.333	2.382	2.379	2.254	2.245	2.306	2.397	2.505	2.490	2.390	
	Ebn at Noon	855	898	916	910	895	853	843	848	849	842	801	797	
	Edh at Noon	73	97	108	113	117	133	133	121	101	79	67	66	
All-Sky Solar Radiation	RadAvg	1.40	2.29	3.50	4.34	4.93	5.46	5.46	4.85	3.92	2.27	1.38	0.98	
	RadStd	0.19	0.28	0.26	0.49	0.54	0.42	0.46	0.28	0.32	0.22	0.18	0.09	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+58	N/A

Nomenclature: See separate page